Premnath Saravanan

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SUMMARY

Systems software engineer with experience in low-level systems programming, Linux platforms and proficient in C, C++ with hands-on experience in driver development, kernel internals, system debugging and Python based testing automation. Strong track record of building reliable, performance-focused software for system-level environments.

EDUCATION

- **Master of Science in Electrical and Computer Engineering -** Specialized in Computer Systems and Software.
  Northeastern University, Boston, MA, USA. Aug 2021 – Dec 2023
- **Bachelor of Engineering in Electrical and Electronics Engineering.**
  RMD College of Engineering - Anna University, Chennai, India. May 2018

SKILLS

- ***Languages and Operating Systems*:** C, C++, Python, Shell scripting, Linux, FreeBSD, RTOS.
- **Systems & Kernel:** Device Drivers, Kernel Internals, Linux Kernel Modules, Interrupt Handling, Concurrency, Synchronization, Memory Management, Crash Dump Analysis, Performance Optimization.
- **Debugging & Validation:** GDB, KGDB, CoreGDB, Wireshark, tcpdump, Root Cause Analysis, Unit Testing.
- **Tools & Platforms:** Git, CI/CD, Redis, CxxTest, HAL-based Mocking.
- **Networking & Hardware:** TCP/IP, NIC Drivers, DMA, I2C, SPI, UART.

PROFESSIONAL EXPERIENCE

NetApp – RTP, NC, USA **Sep 2024 – Present**

Software Engineer

- Implemented interrupt coalescing tuning logic in the NIC driver layer (C/C++), reducing CPU interrupt overhead by ~8% under sustained high-throughput workloads on ONTAP storage controllers.
- Integrated NVIDIA Spectrum-X Ethernet platform support into ONTAP, leveraging SuperNIC capabilities such as RoCE V2, adaptive routing, and advanced congestion control to achieve better bandwidth utilization for AI/ML cluster workloads.
- Developed NIC hotplug support, enabling live network adapter insertion and replacement without cluster downtime, and improving hardware serviceability across storage nodes.
- Diagnosed and resolved kernel panics, memory leaks, and race conditions using KGDB, CoreGDB, and crash dump analysis; identified root causes and implemented reference-counting safeguards and synchronization fixes.
- Implemented ioctl handlers in the driver layer to expose per-port statistics to user space, enabling real-time visibility into port-level counters for diagnostics and performance monitoring.
- Resolved hotplug regression that caused threads to be preempted during critical teardown and reinitialization paths; worked with partner teams to apply priority-based scheduling for driver work queues and restore stable device recovery.
- Developed CxxTest-based automated unit tests for NIC driver stability validation, covering stress scenarios including port failover, buffer overflow, and counter validation, through HAL-based mocking, improving code coverage by ~20%.

Nokia – Boston, MA, USA **Jan 2023 – Jul 2023**

Embedded Software Developer

- Implemented kernel-level bulk-get path to reduce control-plane lookup latency by 250 ms, improving system responsiveness under load; benchmarked against baseline and validated with soak tests.
- Revamped the database module to enable real-time updates in the Redis database when hardware components were added or removed, significantly enhancing application stability and ensuring data accuracy.
- Collaborated on watchdog timer integration and reliability mechanisms to strengthen fault detection and recovery behavior.
- Tracked and eliminated memory leaks across key modules using Valgrind, improving system stability and reducing crash rates under extended stress and soak testing.

Infosys – Bangalore, India **May 2018 – Jul 2021**

Software Engineer

- Designed and implemented a new feature for parallel job execution by collaborating with business analysts and senior developers, accelerating end-to-end runtime by 27% by eliminating job queueing delays and database read/write contention.
- Resolved performance bottlenecks in a high traffic database by optimizing database queries and implementing caching strategies, resulting in 19% improvement in data loading.
- Collaborated effectively with testing team and developed a custom data setup tool for application testing across various scenarios.

PROJECTS

GPU Cache Memory Analysis (C++, CUDA, GPGPU-Sim)

- Analyzed cache memory using GPGPU simulator with diverse multithreaded workloads across various cache and system configurations. Concentrated on performance improvement and accuracy of processing by working with kernels and schedulers.

3-D Mapping (Python, ROS, SLAM)

- Developed python driver, integrated LiDAR, IMU and GPS using ROS for mapping the real world using an autonomous car. Calibrated IMU and LiDAR point cloud data for better loop closures. Analyzed and visualized the trajectory and point cloud data for mapping.